

Johan Achy Yapo

johan.achy.yapo@gmail.com | [LinkedIn](#) | <https://github.com/johan-yapo>

TECHNICAL SKILLS

Languages: Python, Bash, Java, SQL

Libraries & APIs: Scikit-Learn, Pytorch, TensorFlow, pandas, NumPy, matplotlib, SHapley Additive exPlanations (SHAP), Matminer, Pymatgen, OriginPlot, RESTful

Computational Chemistry: Density Functional Theory (DFT), Spin-Orbit Coupling (SOC), Surface Slab Modeling, Band Structure Analysis, Bader Charge Analysis, Crystal Orbital Hamilton Population (COHP)

Data Science: Data Modeling, Feature Engineering, Recursive Feature Elimination, Hyperparameter Tuning, Data Visualization tools

Software/Tools: VASP, Gaussian, LOBSTER COHP, FullProf Suite, SLURM, Conda, Git, Github, Globus Data Transfer, Jupyter, Spyder

Experimental: Arc-melting, X-ray diffraction (XRD), FT-IR, X-ray fluorescence, GC-MS, UV-VIS spectroscopy, colorimetry, NMR, Glove box operation.

EDUCATION

Ph.D. in Physical Chemistry, University of California, Riverside (UCR) June 2025

Dissertation title: “*Application of Machine Learning Towards Discovering Magnetic Borides*”

Committee members: Boniface Fokwa (chair), Timothy Su, and Hill Harman

B.Sc. in Specialized Chemistry, University of Illinois Urbana-Champaign May 2017

RESEARCH EXPERIENCE

Application of Machine Learning Toward Discovering Magnetic Borides, UCR May 2022- December 2024

- Led an original research project applying machine learning models (Random Forest, XGBoost) to predict intermetallic compounds' magnetic moments using curated Materials Project and JARVIS datasets.
- Developed automated data pipelines for feature engineering, data cleaning, and high-throughput screening of 60,000+ theoretical compounds to accelerate magnetic materials discovery.
- Conducted interpretability analyses using SHAP plots and recursive feature elimination to identify key chemical and structural features influencing magnetic properties.

Vacancy Engineering in the $V_{1-x}Mo_xB_2$ Solid Solution Series for Enhanced Hydrogen Evolution Reaction (HER) Performance August 2021 – May 2022

- Investigated the $V_{1-x}Mo_xB_2$ ($x = 0-1$) solid solution series for hydrogen evolution reaction (HER) electrocatalysis.
- Discovered a volcano trend in HER activity and correlated it to metal-site vacancies and electronic structure changes via DFT calculations.
- Identified $V_{0.3}Mo_{0.7}B_2$ as the composition with optimal HER performance, linking structure-property relationships to catalytic enhancement.
- Demonstrated the potential of vacancy engineering as a design strategy for high-efficiency HER electrocatalysts.

HER Electrocatalysis of Layered $Ni_{n+1}ZnB_n$ ($n=1,2$) MAB Phases January 2021 – March 2021

- Performed DFT calculations identifying nickel basal planes and hollow sites as the most active HER sites, with Gibbs free energy of hydrogen adsorption values comparable to Pt(111) surfaces.
- Established structure–property relationships between basal plane exposure, B–B bonding motifs, and electrocatalytic performance, providing a new design approach for HER catalysts based on MAB phases.

Crystalline CoB as a Highly Active Electrocatalyst for Hydrogen Evolution Reaction August 2020 - December 2020

- Performed DFT calculations to analyze the electronic structure of CoB, revealing favorable hydrogen adsorption energetics and a lack of detrimental antibonding interactions.
- Established a structure-activity relationship connecting the well-ordered crystalline lattice of CoB to its enhanced catalytic behavior.
- Highlighted CoB as a cost-effective, earth-abundant alternative to platinum-based HER catalysts for renewable energy applications.

WORK EXPERIENCE

Course Developer and Instructor, Office of Diversity, Equity, and Inclusion, UCR January 2024 – June 2024

- Demonstrated leadership and collaboration skills as a DEI practitioner, advancing STEM accessibility through the application of inclusive practices and cultural competence.
- Worked as part of a cross-functional team with the UCR School of Education and Office of DEI to develop curriculum and content for a new online DEI course tailored to international graduate students.
- Developed and delivered instructional materials to teach crucial DEI concepts to students with minimal prior experience, strengthening communication and educational outreach skills.

Teaching Assistant, UCR September 2020 – August 2024

- Created tailored lesson plans, lectures, and in-class activities for General Chemistry courses, emphasizing inclusive pedagogical practices to support first-generation college students.
- Demonstrated leadership and cross-functional collaboration skills as a DEI practitioner, advancing STEM accessibility and fostering an equitable research environment.
- Developed and implemented a rubric for teaching assistants to promote inclusive practices and equitable participation in introductory STEM laboratory courses, resulting in improved student engagement and achievement.

Quality Control Technician II, Chemtool Incorporated June 2017 – August 2019

- Conducted over 25 chemical and mechanical quality control tests on grease and lubricant products, including X-ray fluorescence (XRF) analysis, viscosity, and water content testing.
- Calibrated and maintained XRF spectrometers and developed sensitivity curves to ensure data accuracy for raw materials and final products.
- Analyzed test results to recommend real-time formulation adjustments to manufacturing teams, supporting product compliance and continuous process improvement.

PUBLICATIONS

Lee, E., **Yapo, J. A.**, Bhupathy, A., Mentzel, T., Yin, X., & Fokwa, B. P. T. (2025) Volcano-Like Diboride Solid Solutions: Synergistic Antibonding and Vacancy Effects for Superior High-Current-Density Hydrogen Evolution. *Angewandte Chemie*. **(co-first author)** [In submission]

Lee, E., Yapo, J. A. & Fokwa, B. P. T. (2025) Dual-Active Sites and Vacancies in a Volcano-Like Solid Solution with High Current Density Performance for Hydrogen Evolution. **(co-first author)** [In submission]

Rezaie, A. A., Lee, E., Luong, D., **Yapo, J. A.**, & Fokwa, B. P. T. (2022). Fe₅Ge₂Te₂: Iron-rich Layered Chalcogenide for Highly Efficient Hydrogen Evolution. *Zeitschrift für anorganische und allgemeine Chemie*, 648(10), e202200066. <https://doi.org/10.1002/zaac.202200066> **(co-first author)**

Rezaie, A. A., Lee, E., **Yapo, J. A.**, & Fokwa, B. P. T. (2021). Highly Active and Abundant MAB Phases $\text{Ni}_{n+1}\text{ZnB}_n$ ($n=1,2$) Toward Hydrogen Evolution. *Advanced Energy and Sustainability Research*, 2(5), 2100052. <https://doi.org/10.1002/aesr.202100052> (co-first author)

Rezaie, A. A., Lee, E., Luong, D., **Yapo, J. A.**, & Fokwa, B. P. T. (2021). Abundant Active Sites on the Basal Plane and Edges of Layered van der Waals Fe_3GeTe_2 for Highly Efficient Hydrogen Evolution. *ACS Materials Letters*, 3(3), 313–319. <https://doi.org/10.1021/acsmaterialslett.1c00048>

CERTIFICATIONS

Chancellor’s “Making Excellence Inclusive” DEI Certificate Program, UCR
IBM Data Science by IBM on Coursera
SQL Essentials on LinkedIn

June 2023
March 2025 - Currently
April 2025

AWARDS

Alliances for Graduate Education and the Professoriate (AGEP) NSF Funding Award
Advancing Faculty Diversity Award, UCR Department of Chemistry
President’s Award Program, University of Illinois Urbana-Champaign
Liberals Arts and Sciences Achievement Award

January 2020– January 2021
September 2024 – December 2024
August 2012– May 2017
August 2015 – May 2017